

CS-AML

Investigation Methodology

Version 0.1.4

Approved Internal Specification Baseline

2026-10-10 · tag v0.1.4-spec

Civil Society Anti-Money Laundering & Financial Intelligence Framework

Generated from the canonical Markdown source. The Markdown file in the repository remains authoritative; change tags such as `[v0.1.x · ...]` are part of the traceability record.

Contents

0. Document Control	5
1. Purpose, Status, and Methodological Objectives	6
1.1 Primary methodological goal	6
1.2 Operational objectives	6
1.3 Methodological boundaries	6
2. Foundations and External Alignment	6
3. Investigation Lifecycle	7
4. Stage 1 — Intake and Triage	7
4.1 Intake sources	7
4.2 Intake record requirements	7
4.3 Triage questions	8
4.4 Intake rejection criteria	8
5. Stage 2 — Investigation Charter and Question Design	8
5.1 Investigation question	8
5.2 Charter fields	8
5.3 Scope change control	9
6. Stage 3 — Investigation Risk and Harm Assessment	9
6.1 Investigation risk rating	9
7. Stage 4 — Collection Planning	9
7.1 Collection priority model	10
8. Stage 5 — Lawful Collection Methodology	10
8.1 Source classes	10
8.2 Collection record	10
8.3 Web preservation	11
9. Source Evaluation and Provenance	11
9.1 Provenance requirement	11
10. Evidence Preservation and Integrity	11
10.1 Evidence object minimum fields	11
10.2 Original and derivative separation	12
10.3 Chain-of-custody-lite	12
11. Claim, Fact, and Proposition Management	12
12. Entity Resolution Methodology	13
12.1 Resolution dimensions	13
12.2 Resolution outcomes	13
12.3 Merge decision record	14
13. Relationship and Control Mapping	14
13.1 Ownership versus control	14
14. Event and Timeline Analysis	15
14.1 Event object	15
14.2 Timeline techniques	15
15. Asset Tracing and Asset Attribution	15
15.1 Asset classes	15
15.2 Attribution states	15

16. Value-Flow Reconstruction	16
16.1 Value-flow record	16
16.2 Reconstruction rule	16
17. Typology Analysis	17
17.1 Typology workflow	17
17.2 Indicator classes	17
17.3 Typology consistency levels	17
18. Hypothesis Generation and Testing	17
18.1 Minimum hypothesis set	18
18.2 Hypothesis matrix	18
18.3 Disconfirming search	18
18.4 Hypothesis status	18
19. Structured Analytical Techniques	19
19.1 Recommended techniques	19
19.2 Prohibited analytical shortcuts	19
20. Confidence and Uncertainty	19
20.1 Confidence statement format	20
20.2 Do not quantify without basis	20
21. Intelligence Gaps and Collection Feedback	20
22. Cross-Case and Network Analysis	20
22.1 Cross-case rules	20
22.2 Network investigation triggers	20
23. Assessment Writing	21
23.1 Key judgements	21
23.2 Language discipline	21
23.3 Mandatory caveats	21
24. Peer Review, Challenge, and Red-Team Review	21
24.1 Peer reviewer questions	21
24.2 Red-team review triggers	22
24.3 Review outcomes	22
25. Dissemination, Referral, and Publication	22
25.1 Dissemination classification	22
25.2 Referral threshold	23
25.3 Publication threshold	23
26. Closure, Monitoring, and Reopening	23
26.1 Closure reasons	23
26.2 Closure record	23
26.3 Reopening triggers	23
27. Quality Assurance and Auditability	24
27.1 Minimum QA checks	24
27.2 Effectiveness metrics	24
28. Technology-Enabling Requirements	24
28.1 Automation and AI	25
29. Roles, Competencies, and Separation of Duties	25
30. Conformance Requirements for Methodology	26
30.1 Methodology conformance levels	26
31. Standard Operating Sequence	26

32. Annex A — Investigation Charter Template	27
33. Annex B — Collection Plan Template	27
34. Annex C — Source & Evidence Evaluation Template	27
35. Annex D — Entity Resolution Decision Template	28
36. Annex E — Value-Flow Worksheet	28
37. Annex F — Typology Worksheet	30
38. Annex G — Hypothesis Matrix Template	30
39. Annex H — Intelligence Gap Register	30
40. Annex I — Peer Review Checklist	30
41. Annex J — Intelligence Product Template	31
42. Annex K — Closure / Monitoring Record	31
43. Glossary	31
44. References and Source Lineage	32

CS-AML INVESTIGATION METHODOLOGY

Derived Methodology · Version 0.1.4

Civil Society Anti-Money Laundering & Financial Intelligence Framework

Approved Internal Specification Baseline · October 2026

Civil Society Anti-Money Laundering & Financial Intelligence Framework — Derived Methodology

Status: Approved Internal Specification Baseline (2026-10-10, tag v0.1.4-spec) | Version: 0.1.4 | Date: 2026-10-10 [v0.1.1 · A01]

Document status — v0.1.4

Version: 0.1.4 — Approved Internal Specification Baseline (2026-10-10, tag v0.1.4-spec). Supersedes v0.1.2 (2026-10-09, tag v0.1.2-spec); unchanged in v0.1.3. [v0.1.4] Supersedes: CS-AML Investigation Methodology v0.1. The DOCX/PDF files in this repository are the unchanged v0.1 baseline (legacy); this Markdown file is the canonical source.

Validation: approved by the product owner as the internal specification baseline on 2026-10-08 (v0.1.1) and 2026-10-09 (v0.1.2) and 2026-10-10 (v0.1.4; decision register and release gates in CHANGELOG.md). The v0.1.2 and v0.1.4 changes come from change requests raised while implementing increments I1-I4 and the post-MVP follow-ups; this is not an independent audit. Acceptance criteria in this document are targets, not evidence that tests have passed.

CS-AML is not an external standard or certification. References to FATF, Wolfsberg, PPATK, UNODC or other bodies do not imply their endorsement.

Changes in 0.1.1: see CHANGELOG.md at the repository root (audit findings A01-A16).

Changes in 0.1.2: change requests CR-I1-01...CR-I4-14 approved by the product owner on 2026-10-09 (CHANGELOG.md, section v0.1.2). Each change is tagged *[v0.1.2 · CR-xx-yy]*.

Changes in 0.1.4: change requests CR-N-01...CR-N-14 approved by the product owner on 2026-10-10 (CHANGELOG.md, section v0.1.4). Each change is tagged *[v0.1.4 · CR-N-xx]*.

0. Document Control

Field	Value
Document title	CS-AML Investigation Methodology v0.1.4 [v0.1.4]
Parent standard	CS-AML Framework v0.1.4 Expanded (CS-AML_Framework_v0.1.4_Expanded.md) [v0.1.1 · A01]
Companion standard	CS-AML Typology Catalogue v0.1.2 (CS-AML_Typology_Catalogue_v0.1.2.md) [v0.1.1 · A01]
Status	Approved Internal Specification Baseline (2026-10-10, tag v0.1.4-spec) / Derived Methodology [v0.1.1 · A01]
Primary audience	Civil society organisations, investigative journalists, public-interest researchers, anti-corruption organisations, environmental and human-rights organisations, digital-rights organisations, research institutions, and trusted technical partners.
Primary use	Lawful, evidence-based financial investigation and intelligence analysis performed without coercive state powers or privileged access to regulated financial systems.
Normative language	MUST/SHALL = mandatory; SHOULD = recommended unless documented justification exists; MAY = optional capability.

This methodology specifies HOW a CS-AML investigation SHALL be conducted. It operationalises the parent framework's analytical chain and translates the Typology Catalogue into a controlled case workflow.

Normative analytical chain: SOURCE → EVIDENCE → CLAIM → FACT → INDICATOR → HYPOTHESIS → ASSESSMENT → INTELLIGENCE PRODUCT

A material allegation, referral, or publication decision SHALL NOT bypass the stages needed to demonstrate provenance, corroboration, competing explanations, confidence, and uncertainty.

1. Purpose, Status, and Methodological Objectives

The purpose of this methodology is to create a repeatable, reviewable, auditable, and rights-respecting method for civil society financial investigations. It is designed for investigations where analysts may have rich public-source, corporate, procurement, legal, property, digital, or documentary information but do not possess compulsory powers, bank secrecy overrides, subpoena authority, or routine access to account-level financial records.

1.1 Primary methodological goal

The primary goal is to convert lawfully obtained information into decision-useful financial intelligence without converting suspicion into accusation prematurely.

1.2 Operational objectives

- Frame investigations around answerable questions rather than predetermined guilt narratives.
- Prioritise collection according to analytical value, legality, proportionality, and risk.
- Preserve source provenance and evidence integrity sufficiently for peer review and referral.
- Resolve identities and relationships before relying on network patterns.
- Reconstruct economic value flows when direct transaction data are unavailable.
- Use AML typologies as analytical lenses rather than proof templates.
- Test competing hypotheses and actively search for disconfirming evidence.
- State confidence, uncertainty, and intelligence gaps explicitly.
- Separate internal analytical judgements from externally publishable allegations.
- Create outputs that can be understood and re-evaluated by another qualified analyst.

1.3 Methodological boundaries

This methodology does not authorise hacking, covert interception, credential theft, deception to obtain protected financial data, unlawful processing of personal data, automated guilt scoring, or the representation of civil society analysis as an official FIU or law-enforcement determination.

2. Foundations and External Alignment

CS-AML adopts a risk-based and evidence-based approach. FATF standards treat understanding of money-laundering risk as central to prioritising resources. FATF financial-investigation guidance recognises financial investigation and asset tracing as core operational elements for competent authorities (pending verification — specific publication/section not yet identified), while UNODC recognises that civil society can support asset tracing through open-source investigation, financial investigation, forensic auditing, and legal analysis (pending verification — page-level support not confirmed). [v0.1.1 · A13] PPATK has publicly recognised that information from NGO/CSO and the public can support early detection and financial-intelligence analysis. CS-AML adapts these ideas to a non-coercive civil-society setting.

Reference lineage	Methodological implication
FATF Recommendations (amended June 2026)	Risk-based proportionality; focus resources on higher-risk areas; distinguish technical controls from effectiveness.
FATF Money Laundering National Risk Assessment Guidance (updated 2025)	Use structured, evidence-based, dynamic risk understanding; document assumptions and changing context.
FATF Financial Investigations Guidance (pending verification — specific publication/section not yet identified)	Treat financial investigation and asset tracing as structured operational disciplines; maintain links between evidence, proceeds, assets, and offences. [v0.1.1 · A13]
UNODC asset tracing / civil-society guidance (pending verification — page-level support not confirmed)	Use public records and lawful sources; generate information on assets, origin, ownership, and control; distinguish access authority from analytical usefulness. [v0.1.1 · A13]
PPATK NGO/CSO public complaint initiative (2025)	Improve quality, structure, and supporting information in civil-society referrals.

3. Investigation Lifecycle

Standard lifecycle: INTAKE → TRIAGE → CHARTER → RISK REVIEW → COLLECTION PLAN → COLLECTION → STRUCTURE → ANALYSIS → HYPOTHESIS TESTING → ASSESSMENT → REVIEW → DISSEMINATION → CLOSURE / MONITORING

The lifecycle is iterative. Analysts MAY return to earlier stages when new information changes scope or hypotheses. However, material decisions SHALL pass the applicable control gate before proceeding.

Gate	Decision	Minimum evidence of readiness
G0 Intake legitimacy	Is the matter suitable for CS-AML work?	Trigger, public-interest rationale, preliminary legal/safety concerns, no prohibited purpose.
G1 Charter approval	Is the investigation question bounded and proportionate?	Approved charter, scope, subjects, period, jurisdictions, initial hypotheses, expected outputs.
G2 Collection approval	May planned data be collected and retained?	Collection plan, data classes, lawful basis/access route, necessity, safeguards, source-risk assessment.
G3 Analytical readiness	Is the case sufficiently structured for material analysis?	Core entities resolved, provenance linked, source conflicts logged, major gaps identified.
G4 Assessment quality	Is the judgement analytically supportable?	Supporting and contradicting evidence, alternatives, typology analysis, confidence, gaps, assumptions.
G5 Dissemination approval	Can the product be shared, referred, or published?	Peer review, legal/privacy review where needed, handling classification, redactions, approval.
G6 Closure/monitoring	Can active investigation stop or change status?	Outcome, unresolved gaps, retention decision, monitoring rationale, lessons learned.

System support for the gates (which criteria are checked as data, which remain reviewer judgement, and the gate → case-status mapping) is stated in Framework §5.9. Converting a case to monitoring and reopening it (§26) are recorded outside the case status in v0.1.x; system support is deferred to v0.2. [v0.1.4 · CR-N-03, CR-N-05]

4. Stage 1 — Intake and Triage

4.1 Intake sources

A case may originate from a whistleblower report, media investigation, public database anomaly, procurement analysis, court record, partner referral, internal research lead, leaked material lawfully received, or prior-case correlation. Intake source type does not determine truthfulness.

4.2 Intake record requirements

- Unique intake identifier and date/time.

- Origin and channel.
- Raw allegation or question preserved separately from analyst interpretation.
- Named subjects or entities, if any.
- Immediate safety, privacy, source-protection, or legal concerns.
- Initial public-interest rationale.
- Initial data sensitivity classification.
- Disposition: reject, hold, preliminary research, or open case.

4.3 Triage questions

Question	Why it matters
What exactly is being alleged or questioned?	Prevents scope drift and conflation of separate allegations.
What would make the issue a financial-intelligence matter?	Ensures there is an ownership, asset, contract, value-flow, proceeds, or control dimension.
What is already known from credible sources?	Avoids duplicating basic validation work.
What is the potential harm if wrong?	Drives proportionality, review, and publication thresholds.
Is there a plausible lawful path to useful evidence?	Prevents investigations that would depend on prohibited access.
Is there an urgent preservation need?	Identifies disappearing web pages, documents, source risk, or volatile digital material.

4.4 Intake rejection criteria

A matter SHOULD be rejected or deferred when it is primarily based on protected characteristics, partisan retaliation, personal vendetta, speculative association, absence of public-interest rationale, impossible lawful collection, or demands for unlawful surveillance. Rejection SHALL be documented without destroying the original intake record if retention is justified.

5. Stage 2 — Investigation Charter and Question Design

Every Investigation or Network Investigation SHALL have an Investigation Charter. The charter is the control against open-ended surveillance and confirmation bias.

5.1 Investigation question

The investigation question SHALL be answerable through evidence and SHALL NOT presuppose criminal guilt. Preferred formulations ask who controls, who benefits, what changed, where value moved, whether declared and observed structures are consistent, and what explanations fit the evidence.

Poor question: “How is Person X laundering money?”

Preferred question: “What ownership, control, asset, contract, and value-flow relationships link Person X and the identified entities during the defined period, and which legitimate or illicit explanations are consistent with the evidence?”

5.2 Charter fields

- Case ID and title.
- Public-interest purpose.

- Primary and secondary investigation questions.
- Subjects and excluded subjects.
- Time period.
- Jurisdictions.
- Predicate issue(s), if any, described as alleged or established.
- Initial hypotheses, including at least one non-criminal alternative when plausible.
- Expected data classes and high-risk data classes.
- Expected outputs and intended audiences.
- Risk classification and required reviewers.
- Stop conditions and review date.

5.3 Scope change control

Material expansion to new persons, jurisdictions, sensitive data classes, or alleged predicate offences SHALL be documented as a scope amendment and reassessed for necessity, legal risk, and proportionality.

6. Stage 3 — Investigation Risk and Harm Assessment

Risk assessment in CS-AML concerns both the subject matter and the conduct of the investigation. It SHALL NOT be reduced to “risk of money laundering”.

Risk domain	Examples	Required response
Legal/regulatory	Privacy, defamation, secrecy restrictions, court orders, cross-border data law	Legal review, narrower scope, controlled access, jurisdictional analysis.
Source safety	Whistleblower exposure, retaliation, coercion	Need-to-know identity access, secure channels, source-risk plan.
Subject harm	Reputational harm, vulnerable persons, mistaken identity	Higher verification threshold, minimisation, redaction, right-of-reply review.
Analyst safety	Harassment, travel risk, digital targeting	Operational security controls, role separation, incident plan.
Data security	Sensitive financial/personal data, credentials, leaked datasets	Restricted repository, encryption, logging, retention limits.
Analytical risk	Confirmation bias, graph overreach, weak source base	Competing hypotheses, peer challenge, confidence discipline.
Publication/referral risk	Irreversible disclosure, source compromise	Approval gate, legal/privacy review, dissemination controls.

6.1 Investigation risk rating

Organisations SHOULD classify investigation risk as Low, Moderate, High, or Critical. The rating SHALL determine review intensity, access controls, approval level, and whether specialist legal/security advice is required. Risk rating SHALL be revisited after major new evidence or scope expansion.

7. Stage 4 — Collection Planning

Collection SHALL be driven by analytical requirements, not by curiosity or tool availability. The Collection Plan links each information requirement to a lawful source strategy.

Collection-plan field	Required content
Information Requirement (IR)	Specific question whose answer changes an analytical judgement.
Priority	Critical / High / Normal / Low.
Target entity/event/value flow	What the IR concerns.
Expected source classes	Registries, court files, procurement, corporate filings, web, interviews, documents, etc.
Collection method	Manual search, API, archive, interview, partner request, document review.
Lawful access note	Why the method is permitted and proportionate.
Sensitivity	Expected personal/confidential/high-risk data.
Success condition	What constitutes sufficient answer.
Stop condition	When additional collection would add little value or increase harm disproportionately.

7.1 Collection priority model

Priority SHOULD consider analytical value × urgency × source volatility × risk reduction. High-volume collection is not inherently better. An organisation SHOULD prefer the smallest set of sources that can answer the question with adequate corroboration.

8. Stage 5 — Lawful Collection Methodology

8.1 Source classes

Class	Examples	Default posture
Authoritative public records	Court decisions, government registries, procurement, official disclosures	High utility; verify currency, scope, self-reporting limitations.
Corporate/public disclosures	Annual reports, websites, ownership statements, filings	Useful; distinguish self-assertion from independently verified fact.
Professional journalism / research	Investigative reporting, NGO reports, academic work	Use as lead and secondary corroboration; inspect underlying evidence where possible.
Open web / social media	Public profiles, posts, archived pages	Volatile; preserve context and avoid identity inference from weak signals.
Human sources	Whistleblowers, witnesses, experts	Record access restrictions, reliability, corroboration, and source-protection controls.
Lawfully received non-public documents	Contracts, invoices, statements, correspondence	Verify authenticity and access basis; restrict handling.
Leaked/breached datasets	Public leak or provided dataset	High legal/security risk; require review before ingestion or redistribution.
Prohibited acquisition	Hacking, stolen credentials, covert interception, impersonation	MUST NOT be used as collection methods.

8.2 Collection record

- Source ID.
- Collector.
- Date/time and method.
- Original URL/location or providing source.
- Access conditions and licence/terms where relevant.
- Original file or snapshot hash where material.
- Sensitivity and handling marking.

- Preservation action.
- Legal/access note.
- Related Information Requirements.

8.3 Web preservation

Material web evidence SHOULD be preserved with URL, access time, full-page or document capture where appropriate, contextual metadata, and hash. Screenshots alone are weak preservation when underlying HTML/PDF/downloadable records are available.

9. Source Evaluation and Provenance

The methodology separates source reliability from information credibility. A reliable institution can publish incomplete or self-reported data; an unknown source can provide a document later independently verified.

Source reliability	Meaning
A	Highly reliable
B	Generally reliable
C	Mixed or variable
D	Generally unreliable
E	Unreliable
F	Unknown / cannot yet be assessed

Information credibility	Meaning
1	Independently confirmed / directly supported
2	Probably true; strong corroboration
3	Possibly true; plausible but incomplete
4	Doubtful; material conflicts or weak support
5	Improbable
6	Cannot yet be assessed

Ratings SHALL be justified in notes for material evidence. A rating such as A1 or F3 is an analytical aid, not a mathematical truth.

9.1 Provenance requirement

Every material fact, relationship, event, asset attribution, and value flow SHALL link to at least one evidence item. Derived calculations SHALL record input evidence and transformation logic.

10. Evidence Preservation and Integrity

Civil-society intelligence is not automatically courtroom evidence. Nevertheless, preserving integrity increases credibility, referral value, and reproducibility.

10.1 Evidence object minimum fields

- Evidence ID.
- Source ID.

- Description.
- File/object location.
- Acquisition date/time.
- Original/derivative status.
- Cryptographic hash for material digital files where feasible.
- Extractor/analyst.
- Sensitivity.
- Authentication/verification status.
- Related claims, entities, relationships, events, and hypotheses.

10.2 Original and derivative separation

Original files SHALL be preserved read-only when feasible. Analyst extracts, translations, OCR, annotations, charts, and redacted copies SHALL be stored as derivatives linked to the original. Translations SHALL keep the source-language text next to the translation. [v0.1.4 · CR-N-10]

10.3 Chain-of-custody-lite

Where an evidence item may be referred to authorities or contested, the organisation SHOULD record acquisition, transfers, transformations, access, and integrity checks. This is a civil-society preservation control and does not claim formal evidentiary admissibility.

11. Claim, Fact, and Proposition Management

Analytical discipline requires separating what a source says from what the investigation accepts as established.

State	Definition	Example
Claim	A proposition asserted by a source or person.	"Person A controls Company X."
Corroborated claim	A claim supported by multiple or stronger sources; it remains a claim and is not itself a fact. [v0.1.1 · A10]	Registry + contract signature + official profile point to same role.
Fact	A proposition sufficiently established for the current analytical purpose.	Person A is listed as director of Company X on date Y.
Indicator	A fact/pattern relevant to a typology or risk.	Three related companies share the same address and director.
Hypothesis	A testable explanatory proposition.	Company Y may act as nominee holder for Person A.
Assessment	A reasoned judgement combining evidence, alternatives, confidence, and gaps.	Available evidence moderately supports common control.

Facts SHALL be time-bounded where status may change. "Person A is director" without a date can be materially misleading.

The stored states for these objects follow the Data Model Specification v0.1.4, Sections 7.4–7.6 (approved by product owner, 2026-10-08): a Claim carries `claim_status` (RECORDED, UNDER_REVIEW, CORROBORATED, CONTRADICTED, UNRESOLVED) — "Corroborated claim" above corresponds to CORROBORATED — and a Fact carries `fact_status` (PROVISIONAL, ESTABLISHED, DISPUTED, SUPERSEDED). A Claim never becomes a Fact: it remains a permanent record of what the source asserts. A Fact is a separate object, created with references to the evidence that supports it (`supporting_evidence`, mandatory), optionally the claims that support it (`supporting_claim_refs`), and one or more recorded VerificationDecisions; the creation decision is recorded together with the Fact. [v0.1.1 · C02] [v0.1.1 · A09, A10]

Decision rights: claim VerificationDecisions are recorded by case members with the LEAD, ANALYST or REVIEWER role. A claim moves RECORDED → UNDER_REVIEW → CORROBORATED / CONTRADICTED / UNRESOLVED, and a decided claim MAY return to UNDER_REVIEW for re-review; the decision history is kept. A Fact is ESTABLISHED only by a principal with REVIEWER or LEAD membership on the case who is not the proposer of the Fact. [v0.1.2 · CR-I2-11]

12. Entity Resolution Methodology

Entity resolution SHALL precede high-impact graph conclusions. Name similarity alone is insufficient for merging records.

12.1 Resolution dimensions

- Stable identifiers: registration numbers, national/company identifiers where lawfully available.
- Names and aliases.
- Date/place of birth or incorporation where lawfully available.
- Addresses.
- Phone/email/domain identifiers.
- Directors/shareholders/beneficial owners.
- Shared documents and signatures.
- Temporal consistency.
- Known relationships and operational context.

12.2 Resolution outcomes

Resolution outcomes are recorded as values of ResolutionDecision.decision (Data Model Specification v0.1.4, Section 8.4). They are decisions about a pair or set of records, separate from each entity's resolution_status state (UNRESOLVED, RESOLVED, CONFLICTED, MERGED, SPLIT), which changes only as an effect of these decisions. [v0.1.1 · ER]

Decision	Meaning
MERGE	Records represent the same entity with sufficient confidence. Absorbed records become MERGED; the surviving record becomes RESOLVED. [v0.1.1 · ER]
POSSIBLE_MATCH	Likely or possible same entity; keep separate records with candidate link. No state change. [v0.1.1 · A09, ER]
KEEP_SEPARATE	Evidence indicates distinct entities. No state change; the pair is not re-suggested unless new evidence is attached. [v0.1.1 · ER]
DEFER	Insufficient evidence. Records stay or become UNRESOLVED. [v0.1.1 · ER]
UNMERGE	Reverses an earlier MERGE (reference required); restored records become SPLIT. Because a merge is logical repointing (absorbed records point to the survivor through canonical_parent; relationships, claims and absorbed records are never rewritten), unmerge re-attributes nothing: it removes what the merge added to the survivor and restores its prior handling. [v0.1.1 · ER] [v0.1.2 · CR-I2-13]

Legacy outcome names (v0.1): MERGED → MERGE; LINKED_POSSIBLE → POSSIBLE_MATCH; SEPARATE → KEEP_SEPARATE; UNRESOLVED → DEFER. [v0.1.1 · ER]

12.3 Merge decision record

Every resolution decision, including merge and unmerge, is stored as an append-only Resolution-Decision (Data Model Specification v0.1.4, Section 8.4). A mistaken decision is corrected by a later decision, never by editing the earlier one. The record holds: *[v0.1.1 · ER]*

- Candidate records.
- Matching and conflicting attributes.
- Evidence.
- Analyst decision.
- Confidence.
- Reviewer (required for every MERGE and UNMERGE). *[v0.1.2 · CR-I2-10]*
- Reversal history if later split (an UNMERGE decision referencing the reversed MERGE). *[v0.1.1 · ER]*

A mistaken entity merge can contaminate every downstream relationship. In the MVP every MERGE and UNMERGE is treated as high-impact: it SHALL name a reviewer who holds REVIEWER or LEAD membership on the case of every subject entity and who is not the decider (reviewer = decider is rejected). *[v0.1.2 · CR-I2-10]*

13. Relationship and Control Mapping

Relationships SHALL be represented as typed, dated, evidence-linked propositions. Graph proximity is not equivalent to control or wrongdoing.

Relationship class	Examples
Legal ownership	OWNS (shareholding recorded as OWNS with an OwnershipInterest), BENEFICIAL_OWNER_OF <i>[v0.1.1 · C09]</i>
Governance/control	DIRECTOR_OF, COMMISSIONER_OF, CONTROLS, AUTHORIZED_SIGNATORY_OF <i>[v0.1.1 · C09]</i>
Economic	PAID_BY, CONTRACTED_BY, SUPPLIER_TO, LENDER_TO (a borrower is the to_entity of LENDER_TO), FUNDED_BY, DONATED_TO <i>[v0.1.1 · C09]</i>
Asset	OWNS, USES, LEASED_TO, ACQUIRED, SOLD_TO ("acquired from" is the inverse of SOLD_TO) <i>[v0.1.1 · C09]</i>
Personal/association	RELATED_TO (familial subtype), ASSOCIATE_OF — only when relevant and lawfully supported <i>[v0.1.1 · C09]</i>
Infrastructure	SHARES_ADDRESS_WITH, SHARES_PHONE_WITH, SHARES_DOMAIN_WITH, SHARES_DEVICE_WITH <i>[v0.1.1 · C09]</i>
Transactional/value	TRANSFERRED_VALUE_TO ("received value from" is the inverse) — only with evidence or clearly marked reconstruction <i>[v0.1.1 · C09]</i>

All values above are registered relationship_type wire values (schemas/enums.yaml; Data Model Annex B). Other relationship names (for example SHAREHOLDER_OF, RELATIVE_OF, ACQUIRED_FROM) are display synonyms mapped in Data Model Annex B and SHALL NOT be stored as wire values. *[v0.1.1 · C09]*

13.1 Ownership versus control

Analysts SHALL distinguish legal ownership, beneficial ownership, operational control, economic benefit, and mere use. The framework MUST NOT convert association or use into ownership without evidence.

An inferred ownership interest (ownership_type BENEFICIAL, ECONOMIC_INTEREST or NOMINEE_ASSERTED) SHALL remain asserted and becomes ESTABLISHED only when its supporting evidence includes an ESTABLISHED Fact. *[v0.1.2 · CR-I3-11]*

14. Event and Timeline Analysis

Timeline analysis is mandatory when sequence materially affects interpretation. Temporal coincidence can generate a lead but does not establish causation.

14.1 Event object

- Event ID and type.
- Date or date range with precision flag (`temporal_precision`: DATETIME, DAY, MONTH, YEAR, RANGE, UNKNOWN; the date string SHALL match the precision). An event with an unknown date has no start time and precision UNKNOWN; a missing start time is permitted only with UNKNOWN. *[v0.1.2 · CR-I3-06]*
- Entities involved.
- Location/jurisdiction if relevant.
- Evidence links.
- Status: confirmed / probable / possible / disputed. These are analytical labels, not yet registered wire values: the stored envelope status of an Event is RECORDED, and an event status enum is deferred to v0.2. *[v0.1.2 · CR-I3-05]*
- Analytical relevance.

14.2 Timeline techniques

- Compare appointments, incorporation, procurement awards, payments, loans, property acquisitions, disposals, litigation, and ownership changes.
- Mark known versus estimated dates (approximate dates are flagged, never presented as exact). *[v0.1.2 · CR-I3-06]*
- Identify events that precede or follow value creation.
- Test whether alleged causal narratives remain plausible under the actual sequence.
- Record missing periods and unknown dates as intelligence gaps.

15. Asset Tracing and Asset Attribution

Asset tracing under CS-AML seeks to identify assets, ownership, control, acquisition/disposal, value, and relationships to potential proceeds or value flows using lawful sources.

15.1 Asset classes

- Real property.
- Vehicles, vessels, aircraft.
- Corporate shares and beneficial interests.
- Securities/investments where lawfully observable.
- Crypto-assets/wallets where lawfully attributable.
- Precious metals, art, luxury goods, portable stores of value.
- Intellectual property or contractual rights when economically material.

15.2 Attribution states

State	Meaning
LEGAL_OWNER	Ownership supported by authoritative or strong documentary evidence.
BENEFICIAL_INTEREST	Evidence indicates economic benefit/control distinct from legal title.
CONTROL/USE	Subject appears to control or use asset; ownership unproven.
ASSOCIATED	Asset linked through a related entity; no ownership inference.
UNRESOLVED	Attribution not established.

Observed associated assets SHALL NOT be described as “hidden wealth” unless the evidentiary and legal basis supports that characterisation.

16. Value-Flow Reconstruction

Follow-the-value is the central analytical adaptation for civil society. A value flow SHALL be classified as DIRECT, DOCUMENTED, RECONSTRUCTED, or HYPOTHETICAL (wire values of `flow_class`, defined in the Data Model Specification v0.1.4, Annex A). The method SHALL preserve that distinction. [v0.1.1 · A09]

Flow class	Evidence threshold	Example
DIRECT	Direct transaction/account/blockchain record lawfully available.	Account A → Account B, amount/date supported by record.
DOCUMENTED (display: Documented economic) [v0.1.1 · A09]	Contract, invoice, loan, dividend, asset sale, grant, procurement payment documented.	Agency awards Rp X contract to Company A.
RECONSTRUCTED	Sequence inferred from multiple economic events without direct transfer record.	Contract revenue precedes related entity property acquisition; causal link not directly proven.
HYPOTHETICAL	Analytical scenario requiring evidence.	Possible movement through intermediary Company C.

16.1 Value-flow record

- Origin entity.
- Destination entity or asset.
- Value and currency if known.
- Date/range.
- Mechanism.
- Flow class.
- Evidence links.
- Confidence.
- Assumptions.
- Alternative explanations.

16.2 Reconstruction rule

A reconstructed flow SHALL NOT be visualised or described in a way that is indistinguishable from a directly evidenced transaction. Diagrams SHOULD use different line styles and display labels (e.g. “Documented”, “Reconstructed”, “Hypothetical”) mapped from the stored `flow_class` value; display labels SHALL NOT replace the stored value. [v0.1.1 · A09]

Certainty rules: for invariants only, the classes are ordered HYPOTHETICAL < RECONSTRUCTED < DOCUMENTED < DIRECT (the order is never displayed, averaged or summed). Raising a flow's

class requires at least one evidence reference not previously attached; a chain of legs is never stronger than its weakest leg, each leg keeps its own class, evidence, value and confidence, and legs are immutable once recorded. [v0.1.2 · CR-I3-09, CR-I3-04]

Aggregation: values are totalled only within the same flow class, flow type and currency, never as a grand total. Contract and subcontract value is an obligation, not evidence of settlement; only DIRECT flows evidence settlement. Unknown amounts are counted and never treated as zero, and legs are never added to the total of their own flow. [v0.1.2 · CR-I3-12]

17. Typology Analysis

The CS-AML Typology Catalogue is an analytical reference, not a guilt classifier. A typology match assesses consistency between observed mechanisms and known laundering patterns.

17.1 Typology workflow

| Observed facts → Indicators → Candidate typology → Mechanism test → Alternative explanations → Evidence gaps → Consistency assessment

17.2 Indicator classes

Class	Meaning
M	Mechanism-specific indicator: closely tied to the typology mechanism.
C	Corroborating indicator: increases plausibility but is not distinctive alone.
K	Contextual indicator: provides environment/background.
D	Disconfirming indicator: weakens or contradicts the typology explanation.
G	Gap: missing information necessary to test the mechanism.

17.3 Typology consistency levels

Level	Meaning
No analytical basis	Evidence does not meaningfully support the mechanism.
Weak consistency	Some indicators exist but plausible benign explanations dominate or corroboration is weak.
Plausible consistency	Multiple relevant indicators; mechanism is credible but important gaps remain.
Strong consistency	Mechanism is supported by multiple independent evidence lines and alternatives are materially weaker.
Compelling consistency	Available evidence closely and coherently matches the mechanism; nevertheless this is not a legal finding of money laundering.

Except where direct authoritative evidence establishes the mechanism, a typology SHALL NOT be assessed above Plausible consistency solely on one indicator or one source.

18. Hypothesis Generation and Testing

Hypotheses are explicit, testable explanations. They prevent analysts from treating the first plausible narrative as the conclusion.

18.1 Minimum hypothesis set

For material cases, analysts SHOULD maintain: (1) the principal suspected explanation, (2) at least one legitimate/benign explanation where plausible, and (3) an “insufficient information / alternative actor or mechanism” hypothesis where relevant.

Each hypothesis records its role in this set as `hypothesis_role`: PRINCIPAL (1), ALTERNATIVE_LEGITIMATE (2), ALTERNATIVE_MECHANISM or INSUFFICIENT_INFORMATION (3), together with its stated assumptions. [v0.1.2 · CR-I4-06]

18.2 Hypothesis matrix

Evidence / observation	H1 suspected mechanism	H2 legitimate explanation	H3 alternative mechanism	Notes
Common director across companies	Supports	Neutral	Supports	Not distinctive alone.
No observable operations	Supports	Weakens	Supports	Could reflect holding company.
Direct legitimate commercial contract	Neutral	Supports	Neutral	Need pricing/related-party context.
Sequential asset transfer after contract	Supports	Neutral	Supports	Temporal association, not causation.

Matrix cells are stored with the wire values of `hypothesis_link_effect`: SUPPORTS, CONTRADICTS, NEUTRAL, UNKNOWN. The display word “Weakens” maps to CONTRADICTS. Each cell carries a rationale and an append-only history. [v0.1.2 · CR-I4-07]

18.3 Disconfirming search

Analysts SHALL deliberately search for evidence that would weaken the leading hypothesis. This search and its result SHALL be documented for High/Critical cases.

Disconfirming searches MAY be recorded on a draft or a finalized assessment until the assessment (or a product depending on it) passes review approval; after approval the record is frozen. The requirement of at least one search for a high-impact or adverse assessment is checked at review approval. [v0.1.2 · CR-I4-10]

18.4 Hypothesis status

- OPEN — actively tested.
- SUPPORTED — evidence currently favours the hypothesis.
- WEAKENED — contradicting evidence materially reduces plausibility.
- REJECTED — evidence is inconsistent with the hypothesis.
- INCONCLUSIVE — evidence is insufficient or balanced.

Every status change SHALL carry a rationale. SUPPORTED requires at least one SUPPORTS cell; WEAKENED and REJECTED require at least one CONTRADICTS cell. The competing-hypotheses rule is enforced when an assessment is finalized: finalization is refused (COMPETING_HYPOTHESES_REQUIRED) while a hypothesis in the assessment's scope has no competing hypothesis. [v0.1.2 · CR-I4-08]

Finalized and reviewed are distinct: finalizing an assessment freezes its content, while review (peer review, approval, supersession) is a separate later outcome. [v0.1.2 · CR-I4-09] When a supporting Fact changes, dependent assessments are flagged for re-review, both directly and indirectly (through indicators, typology matches or hypotheses that rely on it); a flagged draft cannot be finalized, and clearing the flag is a review action. [v0.1.2 · CR-I4-11]

19. Structured Analytical Techniques

Analysts MAY use structured techniques provided the method is transparent and does not create false mathematical precision.

19.1 Recommended techniques

- Chronology/timeline analysis.
- Entity-link and ownership mapping.
- Value-flow reconstruction.
- Comparison of declared versus observed relationships.
- Pattern and typology mapping.
- Competing-hypothesis matrix.
- Source cross-validation.
- Peer-group or baseline comparison when a defensible baseline exists.
- Cross-case correlation.
- Gap analysis and collection requirements.

19.2 Prohibited analytical shortcuts

- Treating network centrality as guilt.
- Treating PEP status, religion, ethnicity, nationality, activism, NPO status, or political association as a standalone AML indicator.
- Treating secrecy, privacy, offshore use, cash use, remittance, or crypto use as inherently illicit.
- Using opaque AI scores as final findings.
- Converting a weak match into an entity merge to “complete” the graph.
- Conflating correlation, sequence, opportunity, and causation.

20. Confidence and Uncertainty

Confidence describes the analyst’s confidence in an assessment given evidence quality, consistency, independence, coverage, and unresolved alternatives. It does not describe the probability that a person is guilty.

Confidence	Typical conditions
LOW	Material evidence is limited, conflicting, weakly corroborated, or core identity/value-flow questions remain unresolved.
MODERATE	Multiple evidence lines support the judgement, but important gaps or viable alternatives remain.
HIGH	Multiple independent, strong evidence lines converge; key alternatives have been tested and materially weakened; critical gaps are limited.
INSUFFICIENT_BASIS	A judgement was attempted but the available evidential basis is insufficient to support any confidence level. This is not a level below LOW. [v0.1.1 · A09]

Confidence wire values are HIGH, MODERATE, LOW, and INSUFFICIENT_BASIS (Data Model Specification v0.1.4, Section 14.1 and Annex A). A rationale is mandatory for every value. INSUFFICIENT_BASIS SHALL NOT be converted to LOW, null, zero, or omitted; a missing (null) confidence is permitted

only on drafts where no judgement has yet been made, and a finalized assessment SHALL carry a non-null value. [v0.1.1 · A09]

20.1 Confidence statement format

Required pattern: Assessment + confidence + principal basis + principal caveat.

Example: “Available evidence supports with MODERATE confidence that Companies A and B were under common operational control during 2025, based on overlapping directors, authorised signatories, address infrastructure, and contract execution. Beneficial ownership remains unresolved and no direct financial transfer between the companies has been established.”

20.2 Do not quantify without basis

Percent probabilities SHOULD NOT be used unless the organisation has a validated, documented quantitative method and the data support calibration. Narrative confidence is preferred for ordinary CS-AML work.

21. Intelligence Gaps and Collection Feedback

An intelligence gap is a material unknown that affects interpretation, confidence, or actionability. Gaps SHALL be visible, not hidden in prose.

Gap ID	Question	Impact	Priority	Collection option	Status
G-01	Who beneficially controls Company B?	High — affects ownership hypothesis	Critical	BO registry, filings, contracts, interviews	Open
G-02	What consideration was paid for Property X?	Medium — affects value-flow reconstruction	High	Property record, court file, seller source	Open

New gaps SHOULD update the Collection Plan. Collection SHOULD stop when marginal analytical value becomes lower than the legal, safety, privacy, or resource cost.

22. Cross-Case and Network Analysis

Reusable entity records permit cross-case intelligence, but cross-case correlation increases privacy and inference risk.

22.1 Cross-case rules

- A cross-case link SHALL identify its evidence basis and source cases.
- Case access restrictions SHALL carry into cross-case views where appropriate.
- Analysts SHALL NOT expose a sensitive case merely because a shared entity appears in another case.
- Cross-case correlation MAY create a new Network Investigation when the relationship is analytically material.
- Automated similarity suggestions SHALL be treated as leads until reviewed.

22.2 Network investigation triggers

- Repeated beneficial-owner or nominee pattern across unrelated cases.
- Shared intermediaries, addresses, professionals, wallets, companies, or assets across cases.
- Recurring procurement/vendor pattern.

- Repeated asset-conversion sequence.
- Recurring typology with common infrastructure.

23. Assessment Writing

Assessment writing SHALL be precise, bounded, evidence-linked, and calibrated to confidence.

23.1 Key judgements

A final product SHOULD begin with a small number of Key Judgements. Each judgement SHALL state confidence and material caveats.

23.2 Language discipline

Avoid	Prefer
"X laundered money."	"The observed structure is consistent with [typology] to [level], but direct evidence of laundering has not been established."
"X owns the villa."	"The villa is legally owned by Company Y; available evidence indicates X may exercise control/use. Beneficial ownership is unresolved."
"Money flowed from contract to property."	"The property acquisition followed the contract award and is linked through related entities; the intervening financial transfer is reconstructed, not directly evidenced."
"Suspicious company."	"The company exhibits the following documented indicators..."

23.3 Mandatory caveats

Where applicable, products SHALL disclose that civil society lacks compulsory access to bank records, tax records, beneficial-ownership verification, or other non-public data necessary to establish certain mechanisms.

24. Peer Review, Challenge, and Red-Team Review

Review is a substantive analytical control, not copy-editing.

24.1 Peer reviewer questions

- Can every key judgement be traced to evidence?
- Are identity merges defensible?
- Are facts time-bounded?
- Are graph relationships typed accurately?
- Is a reconstructed flow clearly distinguished from a direct flow?
- Were benign alternatives tested?
- What evidence most strongly contradicts the assessment?
- Does confidence match evidence quality and gaps?
- Is personal data necessary and proportionate?
- Could wording imply guilt beyond the evidence?

24.2 Red-team review triggers

High/Critical cases, major public allegations, cases involving vulnerable persons, high-profile officials, cross-border legal exposure, or novel analytical models **SHOULD** receive enhanced challenge by a reviewer who was not part of the core investigation.

24.3 Review outcomes

- APPROVE.
- APPROVE WITH CONDITIONS.
- RETURN FOR ANALYSIS.
- NARROW OR REDACT.
- DO NOT DISSEMINATE.

25. Dissemination, Referral, and Publication

The same analysis may require different products for internal decision-makers, trusted partners, competent authorities, or public publication. Dissemination **SHALL** be purpose-limited.

Product	Typical audience	Characteristics
Analytical Note	Internal team	Rapid, clearly caveated, may contain unresolved leads.
Investigation Brief	Management/partner	Structured findings, graph/timeline, confidence, gaps.
Financial Intelligence Package	Competent authority / trusted specialist	Evidence index, entity profiles, source provenance, value-flow analysis, typology/hypothesis assessment, handling caveats.
Public Investigation Report	Public/media/advocacy	Fact-checked, minimised personal data, legal/privacy reviewed, source-protection applied.
Referral Memorandum	PPATK/LEA/regulator or authorised recipient	Concise allegation/issue, supporting evidence, identities, relationships, gaps, contact channel, handling restrictions.

25.1 Dissemination classification

Every product **SHALL** carry an information classification from the five-level CS-AML model (Data Model Specification v0.1.4, Section 16): PUBLIC, INTERNAL, SENSITIVE, RESTRICTED, SOURCE_PROTECTED. Unknown or missing classification fails closed. A product inherits the highest classification of its inputs unless a recorded reviewer downgrade decision exists. *[v0.1.1 · A08]*

The intended dissemination scope is recorded separately from the classification, with access labels where needed: *[v0.1.1 · A08]*

- Organisation only (internal use).
- Named project/partner group — recorded as a compartment/purpose access label, not as a classification level.
- Referral — prepared for a competent authority or designated recipient; recorded as a Dissemination record (recipient, purpose), not as a classification level.
- Public release — requires classification PUBLIC after review and approval.

Legacy note: v0.1 listed INTERNAL / RESTRICTED / CONFIDENTIAL / REFERRAL / PUBLIC here. CONFIDENTIAL (need-to-know, sensitive sources or data) maps to SENSITIVE or RESTRICTED as decided by the data owner, or to SOURCE_PROTECTED where the reason is source-identifying information; it **SHALL NOT** be mapped automatically. *[v0.1.1 · A08]*

25.2 Referral threshold

Referral SHOULD be based on actionable, structured information rather than certainty of a crime. A referral SHALL clearly distinguish established facts, analytical assessments, and unresolved allegations. PPATK's public engagement with NGO/CSO emphasises the usefulness of better-quality supporting information; CS-AML therefore prioritises structured, evidence-linked referral packages.

25.3 Publication threshold

Publication normally requires a higher harm-control standard than internal analysis or confidential referral. The organisation SHALL consider factual accuracy, public interest, necessity of naming individuals, source protection, privacy, defamation exposure, and whether the wording fairly reflects uncertainty.

26. Closure, Monitoring, and Reopening

Cases SHALL be closed, suspended, converted to monitoring, or escalated; they SHOULD NOT remain indefinitely open by default.

26.1 Closure reasons

- Question answered.
- Hypothesis rejected or unsupported.
- Insufficient lawful evidence and no proportionate collection path.
- Referred to competent authority or partner.
- Merged into network investigation.
- Risk exceeds organisational capacity.
- Public-interest rationale no longer sufficient.

26.2 Closure record

- Disposition.
- Final key judgement.
- Confidence.
- Unresolved gaps.
- Disseminations/referrals.
- Retention/destruction decision.
- Monitoring triggers, if any.
- Lessons learned and typology updates.

26.3 Reopening triggers

A case MAY reopen on new evidence, authoritative findings, relevant ownership/asset changes, related-case correlation, or a new allegation that materially changes the analytical picture. Reopening SHALL create an audit record and refreshed risk review.

27. Quality Assurance and Auditability

Quality assurance assesses whether the methodology was followed and whether analytical products are useful, proportionate, and reproducible.

27.1 Minimum QA checks

- Case charter completeness.
- Source and provenance completeness.
- Evidence integrity.
- Fact/claim separation.
- Entity-resolution decisions.
- Relationship evidence links.
- Value-flow classification.
- Typology use and caveats.
- Hypothesis testing.
- Confidence calibration.
- Gap visibility.
- Peer review.
- Dissemination approval.
- Retention compliance.

27.2 Effectiveness metrics

Metric family	Examples
Quality	% key judgements with direct evidence links; peer-review rework rate; identity-merge reversal rate.
Timeliness	Time from intake to triage; time to first analytical assessment; referral preparation time.
Actionability	% referrals acknowledged; partner feedback; information requests generated by product.
Proportionality	Volume of sensitive data collected versus used; deletion/minimisation rate.
Learning	Typology updates; recurring gaps; detection of repeated infrastructure across cases.
Safety	Source/security incidents; unauthorized disclosures; high-risk access exceptions.

Metrics SHALL NOT reward volume of suspects, allegations, or personal data collected.

28. Technology-Enabling Requirements

Technology supports methodology; it does not replace analyst judgement. Any CS-AML platform SHOULD implement the following methodological controls.

- Case and scope management.
- Source registry and immutable provenance fields.
- Evidence repository with hashing/versioning.
- Claim/fact/indicator objects.
- Reusable entity store with reversible entity merges.

- Typed, dated, evidence-linked relationships.
- Asset and event models.
- Direct/documented/reconstructed/hypothetical value-flow classes.
- Typology worksheets linked to Catalogue IDs.
- Hypothesis matrix and disconfirming evidence.
- Confidence and intelligence-gap fields.
- Role-based access and sensitive-source compartmentalisation.
- Peer review and approval workflow.
- Audit history.
- Controlled export/redaction.
- Cross-case correlation with access-aware filtering.

28.1 Automation and AI

Automation MAY assist extraction, translation, entity suggestions, document classification, similarity detection, and graph discovery. Automated outputs SHALL be labelled as machine-generated until reviewed. AI SHALL NOT independently establish guilt, assign criminality, merge high-impact identities, or approve dissemination.

- **Offline only.** Every automation engine (OCR, machine translation, any future AI) SHALL run offline on the organisation's own host; evidence content SHALL NOT be sent to a third-party service, and there is no online fallback.
- **Machine translation.** A translation keeps the source text, is labelled "Machine translation — review required" until a person reviews it and remains labelled a translation afterwards. A second (independent) reviewer is not required, but any intelligence product that uses a translation SHALL carry the machine-translation label. A translation is never translated again.
- **Review is not verification.** Reviewing machine text does not create a fact; facts still need a verification decision against evidence (§11). [v0.1.4 · CR-N-10]

29. Roles, Competencies, and Separation of Duties

Role	Core responsibilities	Minimum competency
Case Owner	Purpose, scope, resources, closure	Public-interest rationale; risk judgement; governance.
Lead Analyst	Analysis plan, hypotheses, assessment	Financial investigation concepts; structured analysis; writing.
Collector/Researcher	Lawful acquisition, provenance	OSINT/research methods; source handling.
Evidence Custodian	Integrity, storage, preservation	Digital evidence basics; records management.
Peer Reviewer	Challenge and quality	Independent analytical judgement; framework knowledge.
Legal/Privacy Reviewer	Rights and legal risk	Applicable law/policy; privacy/defamation/data protection.
Security Owner	Operational and information security	Access control, incident response, source protection.
Approver	Dissemination authority	Organisational accountability and risk acceptance.

Small organisations MAY combine roles, but high-impact dissemination SHOULD preserve at least one independent reviewer/approver distinct from the primary analyst.

30. Conformance Requirements for Methodology

An implementation claiming alignment with CS-AML Investigation Methodology v0.1.4 SHALL demonstrate the following minimum artefacts for a material investigation:

- Intake/Triage Record.
- Investigation Charter.
- Investigation Risk Assessment.
- Collection Plan.
- Source Register.
- Evidence Register.
- Entity Resolution Decisions for contested/high-impact identities.
- Relationship/Asset/Event records linked to evidence.
- Value-Flow Worksheet where economic movement is material.
- Typology Worksheet where a typology is referenced.
- Hypothesis Matrix including contradictory/disconfirming evidence.
- Intelligence Gap Register.
- Assessment with confidence statement.
- Peer Review Record.
- Dissemination/Referral Approval.
- Closure or Monitoring Record.
- Audit history sufficient to reconstruct material changes.

30.1 Methodology conformance levels

Level	Characteristics
M1 — Basic	Manual artefacts; minimum provenance, hypothesis, peer review, dissemination approval.
M2 — Operational	Structured registers, formal risk review, entity resolution, typology/value-flow worksheets, QA.
M3 — Integrated	Technology-enforced provenance, graph/value-flow integration, cross-case controls, role-based workflow, metrics.
M4 — Assured	Independent assurance, periodic control testing, calibrated automated analytics where used, mature privacy/security governance.

31. Standard Operating Sequence

- 01 Receive intake
- 02 Preserve original intake
- 03 Triage legitimacy and urgency
- 04 Create Investigation Charter
- 05 Perform risk/harm review
- 06 Define Information Requirements
- 07 Approve Collection Plan
- 08 Collect and preserve sources
- 09 Register evidence and provenance
- 10 Resolve core entities
- 11 Build relationships, assets, and events
- 12 Construct timeline
- 13 Map direct/documented value flows
- 14 Reconstruct value flows where justified

- 15 Map candidate typologies
- 16 Create/test competing hypotheses
- 17 Identify contradictory evidence and gaps
- 18 Draft key judgements and confidence
- 19 Peer/red-team review
- 20 Legal/privacy/publication review where applicable
- 21 Approve dissemination/referral/publication
- 22 Close, monitor, or reopen
- 23 Capture lessons and typology updates

32. Annex A — Investigation Charter Template

Field	Template prompt
Case ID / Title	
Public-interest purpose	Why is this investigation justified?
Primary question	What specific question must be answered?
Secondary questions	
Subjects / entities	
Excluded scope	
Time period	
Jurisdictions	
Predicate issue(s)	Established / alleged / unknown
Initial hypotheses	Include plausible benign alternative
Expected data classes	
Sensitive data expected	
Expected outputs	
Risk level	Low / Moderate / High / Critical
Required reviewers	
Review date	
Stop conditions	

33. Annex B — Collection Plan Template

IR ID	Information Requirement	Priority	Source classes	Method	Lawful-access note	Sensitivity	Success/stop condition
IR-01							
IR-02							
IR-03							

34. Annex C — Source & Evidence Evaluation Template

Field	Entry
Source ID	
Evidence ID	
Source type	
Origin / URL / provider	
Acquisition date/time	
Collection method	
Lawful-access note	
Reliability rating (A-F)	
Information credibility (1-6)	

Field	Entry
Hash / preservation	
Sensitivity / handling	
Related claims	
Verification notes	

35. Annex D — Entity Resolution Decision Template

Field	Entry
Candidate records	
Proposed entity	
Matching attributes	
Conflicting attributes	
Evidence	
Decision	MERGE / POSSIBLE_MATCH / KEEP_SEPARATE / DEFER / UNMERGE (ResolutionDecision.decision; legacy MERGED / LINKED_POSSIBLE / SEPARATE / UNRESOLVED) [v0.1.1 · A09, ER]
Confidence	HIGH / MODERATE / LOW / INSUFFICIENT_BASIS [v0.1.1 · A09]
Analyst	
Reviewer	
Date	
Reversal notes	

36. Annex E — Value-Flow Worksheet

Flow ID	Origin	Destination/asset	Value	Date	Mechanism	Class	Evidence	Confidence	Assumptions/gaps
VF-01									
VF-02									

37. Annex F — Typology Worksheet

Field	Entry
Typology ID / name	
Observed mechanism	
M indicators	
C indicators	
K indicators	
D indicators	
Gaps	
Alternative explanations	
Consistency level	No basis / Weak / Plausible / Strong / Compelling
Confidence / caveat	
Supporting evidence IDs	

38. Annex G — Hypothesis Matrix Template

Evidence / observation	H1	H2	H3	Diagnosticsity / notes
------------------------	----	----	----	------------------------

39. Annex H — Intelligence Gap Register

Gap ID	Question / missing fact	Impact	Priority	Collection option	Owner	Status
G-01						
G-02						

40. Annex I — Peer Review Checklist

- Investigation question remains within approved scope.
- Key facts are evidence-linked and time-bounded.
- Contested entities are resolved appropriately.
- Relationships do not overstate control or ownership.
- Direct and reconstructed value flows are visually/textually distinct.
- Typology analysis includes disconfirming indicators and gaps.
- Leading hypothesis has been challenged.
- Alternative explanations are fairly represented.
- Confidence matches evidence and uncertainty.
- Sensitive personal data are necessary and minimised.
- Key judgements are not stronger than the evidence.
- Referral/publication wording distinguishes fact, assessment, and allegation.

41. Annex J — Intelligence Product Template

- Document control and handling classification.
- Executive assessment / Key Judgements.
- Investigation question and scope.
- Method and limitations.
- Key entities and relationships.
- Timeline.
- Asset analysis.
- Value-flow analysis.
- Typology analysis.
- Hypothesis assessment.
- Contradictory/disconfirming evidence.
- Confidence statement.
- Intelligence gaps.
- Source/evidence index.
- Recommended next steps.
- Dissemination and redaction notes.

42. Annex K — Closure / Monitoring Record

Field	Entry
Case ID	
Disposition	Closed / Suspended / Monitoring / Referred / Merged
Final key judgement	
Confidence	
Unresolved gaps	
Disseminations/referrals	
Retention decision	
Monitoring triggers	
Lessons learned	
Typology/catalogue update needed	
Approved by / date	

43. Glossary

Term	CS-AML meaning
Assessment	A reasoned analytical judgement that states confidence, basis, caveats, and gaps.
Claim	A proposition asserted by a source; not automatically accepted as fact.
Evidence	A preserved item or extract supporting or contradicting a proposition.
Fact	A proposition sufficiently established for the current analytical purpose.
Indicator	A fact or pattern relevant to risk or a typology.
Intelligence gap	A material unknown affecting judgement, confidence, or actionability.
Hypothesis	A testable explanatory proposition.

Term	CS-AML meaning
Provenance	Traceable origin and handling history of information/evidence.
Reconstructed value flow	An inferred economic sequence supported by evidence but lacking a direct transfer record.
Typology	A known pattern or mechanism used to understand possible laundering or financial-crime behaviour.

44. References and Source Lineage

- Financial Action Task Force (FATF), The FATF Recommendations, as amended June 2026.
- FATF, Money Laundering National Risk Assessment Guidance, updated 28 August 2025.
- FATF, Financial Investigations Guidance (Operational Issues). Pending verification — specific publication/section not yet identified. [v0.1.1 · A13]
- FATF, Investigating Professional Money Laundering, Underground Banking, and the Use of Hawala and Other Similar Service Providers, 3 September 2026.
- United Nations Office on Drugs and Crime (UNODC), Civil Society Guide to the UNCAC / civil-society entry points for asset tracing and recovery (https://www.unodc.org/documents/NGO/Corruption/251113-CSU-UNCAC_Guide-Web.pdf). Pending verification — page-level support not confirmed. [v0.1.1 · A13]
- UNODC, Manual on International Cooperation for the Purposes of Confiscation of Proceeds of Crime — asset tracing sections. Pending verification — specific publication/section not yet identified. [v0.1.1 · A13]
- PPAATK, Klinik Dumas Special Edition: PPAATK dan NGO/CSO Perkuat Aduan TPPU melalui peluncuran lapor.ppatk.go.id, 26 November 2025.
- CS-AML Framework v0.1.4 Expanded (CS-AML_Framework_v0.1.4_Expanded.md). [v0.1.1 · A01]
- CS-AML Typology Catalogue v0.1.2 (CS-AML_Typology_Catalogue_v0.1.2.md). [v0.1.1 · A01]
- CS-AML Data Model Specification v0.1.4 (CS-AML_Data_Model_Specification_v0.1.4.md) — authoritative registry for classification, flow-class, and confidence enumerations. [v0.1.1 · A08, A09]

Reference lineage informs the methodology; it does not mean that each CS-AML step, gate (G0–G6), grade, or threshold is derived from a cited source — these are CS-AML design conventions unless a specific source is cited. [v0.1.1 · A13, N01] Reference lineage does not transform CS-AML into an official FATF, UNODC, PPAATK, FIU, law-enforcement, or regulated-entity standard. Jurisdiction-specific legal advice remains necessary for sensitive collection, data processing, referral, and publication decisions.